AdifySkills

n8n Automation Mastery
+91 83838 91716

MENTOR COPY

Module 3 · APIs, Webhooks & Integrations

DAY 10 of 25

# Databases: Postgres, Supabase, Airtable & Notion

"Sheets se aage — real databases ke saath scale karo."

⚡ AAJ HUM YEH BANAYENGE

## Supabase Leads API (Create / Update / Search)

Ek hi webhook se teen kaam: action = create, update ya search. Data Supabase (Postgres) table mein jaata hai aur API JSON lautati hai — apna chhota backend bina server ke.

Postgres · Supabase · Airtable · Notion

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

# 🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

**💡 Yeh lesson kyun zaroori hai?**

Sheets 10,000 rows ke baad slow ho jaati hain. Real clients ko real database chahiye — Postgres/Supabase.

## 🎯 Aaj ke baad aap yeh kar paoge

**01 SQL basics: SELECT, INSERT, UPDATE, JOIN**

🌍 Librarian se: "Delhi wale saare students do" (SELECT), "naya student jodo" (INSERT).
⚙️ SELECT name, phone FROM leads WHERE city = 'Delhi';

**02 Postgres/MySQL nodes & Supabase**

🌍 Bank ka main computer system.
⚙️ Postgres node → Execute Query / Insert / Upsert; Supabase node → rows CRUD.

**03 Airtable & Notion as no-code databases**

🌍 Sundar register (Airtable), digital diary (Notion).
⚙️ Airtable → Create record; Notion → Database page create.

**04 Upsert patterns & deduplication**

🌍 Contact list mein same number do baar save na ho.
⚙️ ON CONFLICT (email) DO UPDATE … / Remove Duplicates node.

**05 n8n Data Tables (built-in storage)**

🌍 Ghar mein chhota locker — bahar jaane ki zarurat nahi.
⚙️ n8n Data Tables: chhote projects ka built-in storage.

**06 Choosing the right DB for a client**

🌍 Chhote kaam ke liye cycle, bade ke liye truck.
⚙️ < 5k rows + non-tech → Airtable; app backend / scale → Postgres/Supabase.

## ⏱ Class ka plan (2 ghante)

| 10m | 35m | 45m | 20m | 10m |
|---|---|---|---|---|

`0:00-0:10` **Recap**

🪄 *Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.*

👩‍🏫 2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

`0:10-0:45` **Concept: Databases**

Sheets vs Airtable vs Postgres

🪄 *Notebook mein 10 naam dhoondhna easy, 10,000 mein mushkil. Database ek super-fast librarian hai.*

👩‍🏫 Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

`0:45-1:30` **Live Build**

Mini CRM on Supabase with n8n API layer

🪄 *Aaj ki kahani yaad hai — 'Library aur smart librarian'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.*

👩‍🏫 Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

`1:30-1:50` **Quiz + Challenge**

Module 3 quiz

🪄 *Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

👩‍🏫 Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

1:50–2:00 **Q&A + Homework**

*🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

👩‍🏫 Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

## 🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

### Table, row, column

Database = Excel jaisa par fast aur reliable.

🪄 **Analogy:** Library ki almirah aur shelves

CONCEPT 02

### SQL basics

SELECT * FROM leads WHERE city = 'Delhi'

🪄 **Analogy:** Librarian se sawaal poochna

CONCEPT 03

### Upsert

Unique key (email) match → update, warna insert.

🪄 **Analogy:** Attendance: naam hai to tick, nahi to naya naam

CONCEPT 04

### No-code DBs

Airtable/Notion — non-tech clients ke liye easy UI.

🪄 **Analogy:** Ready-made furniture

### 🔑 Aaj ka main idea

Jab data bada ho jaata hai (hazaaron rows), Sheets slow aur risky ho jaati hain. Database (Postgres/Supabase) fast, safe aur multi-user hai. Airtable/Notion beech ka raasta hain — non-tech client ke liye easy UI.

- Sheets = notebook, Database = bank ka ledger system
- Airtable = sundar register jismein filters
- Upsert = attendance: naam hai to tick, nahi to add

### 📖 Ek kahani se samjho: Library aur smart librarian

1 📚 **Library = database**
= Postgres / Supabase

2 🗂️ **Almirah = table**
= leads table

3 📖 **Kitaab = row**
= Ek lead record

4 🙋 **Librarian se sawaal**
= SQL SELECT

5 🔁 **Kitaab hai to update, nahi to nayi**
= Upsert

🪄 Ek line mein: “Notebook mein 10 naam dhoondhna easy, 10,000 mein mushkil. Database ek super-fast librarian hai.”

# 🗺️ Aaj ka workflow: Supabase Leads API (Create / Update / Search)

Data left se right chalta hai. Har dabba ek n8n node hai.




**💰 Isse kitna kama sakte ho?**

Startups (mini backend / leads API) — aam taur par **₹15,000 – ₹40,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

## 🛠 Step-by-step build — n8n kholo aur saath chalo

**1** ⚡ **Webhook**
POST /crm/lead

**2** 🗄️ **Supabase**
Upsert into leads (email unique)

**3** ⚡ **Webhook**
GET /crm/search?city=

**4** 🗄️ **Postgres**
SELECT name, phone FROM leads WHERE city = $1

**5** ⚡ **Respond to Webhook**
Results JSON

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

### 🔑 Setup (import ke baad)

- ☐ Connect: **Supabase (project URL + service key)**
- ☐ Supabase SQL editor mein yeh chalao: create table leads (id bigint generated always as identity primary key, name text, email text unique, status text default 'new', created_at timestamptz default now());

### ✅ Test kaise karein

1. curl se action=create bhejo, phir search, phir update
2. Supabase Table Editor mein data dekho

# ⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

**😩 Pehle (manual)**

| | |
|---|---|
| 10,000 rows ki slow Sheet | 10 min |
| Duplicate dhoondhna | 20 min |

30 min

→

**⚡ Ab (n8n)**

DB mein fast search + auto dedupe

0 min

⏱ Search seconds mein, duplicates automatic band.

## ⌨️ Aaj ka key snippet

**Copy karke n8n mein paste karo** — 📋 Select karke copy karo

```
INSERT INTO leads (email, name, city) VALUES ($1, $2, $3)
ON CONFLICT (email) DO UPDATE SET name = EXCLUDED.name, city = EXCLUDED.city;
```

## 📦 Sample data — input se output tak

**📥 Input** — 📋 Select karke copy karo

```
{
  "email": "raj@mail.com",
  "name": "Raj",
  "city": "Jaipur"
}
```

**📤 Output** — 📋 Select karke copy karo

```
[
  {
    "name": "Raj",
    "phone": "9000011111",
    "city": "Jaipur"
  }
]
```

# 💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

**💬 Build it with ChatGPT / Claude** — 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Ek hi webhook se teen kaam: action = create, update ya search. Data Supabase (Postgres) table mein jaata hai aur API JSON lautati hai — apna chhota backend bina server ke.

Flow: Webhook (POST /leads-api) → Switch (action?) → Supabase (create / update / search) → Respond (JSON result)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 **Debug helper prompt** 📋 Select karke copy karo

```
I'm building the n8n workflow "Supabase Leads API (Create / Update / Search)" (nodes: POST /leads-api,
Which Action?, Create Lead, Update Status, Search Leads, Respond).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

## 📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo ( `Ctrl` + `A` , `Ctrl` + `C` ).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
6 nodes · ready to import
adifyskills.com/workflows/day-10-supabase-leads-api.json

🌐 **Lesson page**
Copy buttons, quiz aur video links
adifyskills.com/day/10

## 🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

| Agar yeh ho raha hai… | To yeh karo |
|---|---|
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "Supabase (project URL + service key)" connect karo. |
| **Webhook/Form test par kuch nahi ho raha?** | "Listen for test event" / Test URL use karo aur 2 minute ke andar request bhejo. Production URL sirf Active workflow par chalta hai. |

### ⚠️ Yeh galtiyan sabse common hain

- ! SQL mein user input seedha jodna (SQL injection) — query parameters use karo
- ! Unique constraint ke bina upsert
- ! Supabase service key ko frontend mein expose karna

### ⭐ Pro tips — time bachane wali tricks

- ★ n8n Data Tables chhote projects ke liye built-in storage
- ★ Index lagao frequently searched columns par
- ★ Client ke liye Airtable interface bana do — woh khud data dekh sake

**Phir bhi atke? Koi baat nahi 🙂**
Screenshot + error message ke saath WhatsApp par poochho →

# Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

**🛠 Live lab**

Supabase mein leads table + n8n se create/update/search endpoints.

**📝 Homework**

Notion content calendar ko Airtable ke saath 2-way sync karo.

- [ ] Homework ho gaya

**BEGINNER**

Airtable mein record add

- [ ]

**INTERMEDIATE**

Notion ↔ Airtable sync

- [ ]

**ADVANCED**

Paginated search API with LIMIT/OFFSET

- [ ]

**🧩 Aaj ka challenge — pehle khud try karo**

**🎯 Same email se dobara "create" aaye to error ki jagah "already exists" message lautao.**

**💡 Hint 1**

Pehle search karo, phir decide karo create karna hai ya nahi.

**💡 Hint 2**

Create se pehle Supabase getAll (filter email=eq.X) + IF "item mila?"

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

# 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

**Q1. Data padhne ka SQL command?**

(A) DROP
(B) UPDATE
(C) INSERT
(D) SELECT

**Q2. SQL injection se bachne ke liye?**

(A) Capital letters
(B) Long query
(C) Comments
(D) Query parameters

**Q3. Non-tech client ke liye easy DB?**

(A) Postgres CLI
(B) Airtable
(C) Redis
(D) MongoDB shell

**Q4. Upsert ka matlab?**

(A) Backup
(B) Delete+insert
(C) Update ya insert
(D) Sirf insert

## 📖 Aaj ke naye shabd

**Webhook**
Ek URL jis par koi doosra app data bhejta hai aur aapka workflow turant chal padta hai.

**JSON**
Data likhne ka format: { "naam": "Ravi", "city": "Delhi" } — key aur value ke jode.

**CRM**
Customers aur leads ka record rakhne ka system.

AdifySkills DIGITAL MARKETING INSTITUTE +91 83838 91716

# 👨‍🏫 Mentor notes — class kaise lein

Yeh page sirf aapki copy mein hai. Students ki PDF mein nahi jaata.

- [ ] Supabase free project live banao
- [ ] Sheets vs DB speed comparison 5000 rows par
- [ ] Module 3 quiz

## 🗣️ Har block ke liye mentor tip

**10m — Recap**

2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

Class se poochho: Kal wale workflow mein sabse mushkil step kaunsa laga? · Homework mein kahan error aaya aur kaise theek kiya? · Kal ka ek node naam batao aur woh kya karta hai?

**35m — Concept: Databases**

Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

Class se poochho: Data padhne ka SQL command? · SQL injection se kaise bachein? · Non-tech client ke liye kaunsa DB?

**45m — Live Build**

Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

Class se poochho: Is node ka output agle node mein kaise ja raha hai? · Agar yeh node fail ho to kya hoga? · Isko apne kaam/business mein kahan use kar sakte ho?

**20m — Quiz + Challenge**

Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

Class se poochho: Data padhne ka SQL command? · SQL injection se bachne ke liye? · Non-tech client ke liye easy DB?

**10m — Q&A + Homework**

Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

Class se poochho: Aaj ka sabse useful node kaunsa laga? · Homework mein pehla step kya hoga? · Aaj ki class 1 line mein summarize karo.

## ✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 11

**Lead Generation & CRM Automation**

"Har lead, har baar, 60 seconds ke andar follow-up."

☐ **Day 10 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

AdifySkills DIGITAL MARKETING INSTITUTE +91 83838 91716

# 🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

**Q1 → D**
SELECT

**Q2 → D**
Query parameters

**Q3 → B**
Airtable

**Q4 → C**
Update ya insert

### ✅ Challenge ka solution

Which Action? (create) → Supabase getAll (email=eq.{{ $json.body.email }}, Always Output Data on) → IF {{ $json.id }} exists → Respond "already exists", warna Create Lead.

## 📚 Official docs

- Postgres node ↗
- Supabase node ↗
- Notion node ↗
- Airtable node ↗

## ▶ Videos

- n8n supabase tutorial ↗
- n8n postgres node ↗

**Doubt poochho**
WhatsApp: +91 83838 91716
wa.me/918383891716

**Day 10 lesson page**
Quiz, copy buttons, videos
adifyskills.com/day/10

**+91 83838 91716**
Call / WhatsApp — Adify Skills